

# Predictions of Small-Body Physical Traits Based on Orbital Characteristics

Micro-Project 1

[GitHub Repository](https://github.com/Ethereal-Mining-Company/736D616C6C2D626F64792D7633)

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September 27, 2026

# Problem Statement

Two ways to predict the diameter of a small-body are **(1)** using a combination of very high data collection cost, very high predictive power, and low observational reliability features like albedo (Group 3) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2) and **(2)** using a combination of low data collection cost, high observational reliability features (Group 1) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2)(Ge, 2026; Murcia, 2023; Robinson, 2024). The project looks to determine if the diameter of a small-body can accurately be predicted with only Group 1 features to maintain the low-cost, high-reliability of features that are inexpensive and not variable or prone to error while observing.

# Hypothesis Formulation

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models target diameter (Y) using two feature subsets: an expanded set of low-cost, high-reliability observations with low-cost, medium-reliability observations ( $\mathbf{X}_{full}$ ) and a restricted set of only 'low-cost', 'high reliability' observations ( $\mathbf{X}_{restricted}$ ). The restricted model ( $\mathbf{X}_{restricted}$ ) is preferred due to the lower number and higher reliability of features. Current peer-reviewed research dictates that absolute magnitude (H), a Group 2 feature, is mathematically required to calculate diameter (Ge, 2026).

The hypothesis determines if small-body diameter (Y) can be accurately predicted within a statistical and operational threshold ( $\delta$ ) by the restrictive feature subset compared to the full feature subset, quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{X}_{full})} - R^2_{Model(\mathbf{X}_{restricted})}$$

## Statistical Hypothesis

The statistical hypothesis tests if the restrictive feature subset of Group 1 features maintains a statistically acceptable degree of accuracy in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Statistical Null Hypothesis ( $H_{0,stat}$ ):**  $\Delta R^2_{test} \leq 0$   
The inclusion of Group 2 features yield no improvement or a decrease in the out-of-sample  $\Delta R^2_{test}$ , indicating that the expanded features **do not** increase the accuracy of predicting the target diameter (Y).
- **Statistical Alternative Hypothesis ( $H_{1,stat}$ ):**  $\Delta R^2_{test} > 0$   
The inclusion of Group 2 features yields a statistically significant increase in the out-of-sample  $\Delta R^2_{test}$ , indicating that the expanded features **do** increase the accuracy of predicting the target diameter (Y).

## Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 features maintains an operationally and practically acceptable degree of accuracy ( $\delta$ ) in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Operational Null Hypothesis ( $H_{0,ops}$ ):**  $\Delta R^2_{test} > \delta$   
The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is not** sufficient enough for operational use.
- **Operational Alternative Hypothesis ( $H_{1,ops}$ ):**  $\Delta R^2_{test} \leq \delta$   
The inclusion of Group 2 features does not yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is** sufficient enough for operational use.

# Acquire

In order to test the two-model comparison framework, a comprehensive and reliable dataset is required. The dataset needed to contain characteristics from Group 1 and Group 2 for small-bodies. The [JPL NASA Small-Body Dataset](#) was selected because NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The [JPL NASA Small-Body Dataset](#) contains Group 1 and Group 2 characteristics such as absolute magnitude ( $H$ ), semi-major axis ( $a$ ), eccentricity ( $e$ ), and inclination ( $i$ ) which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.

The [JPL NASA Small-Body Dataset](#) was queried to meet the target variable diameter ( $Y$ ) to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km). This ensures a valid target variable for regression training and the calculation of the coefficient of determination ( $R^2$ ). Based on these parameters, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.

# Prepare

Data has been prepared for regression analysis in the following ways:

1. **Data Structure Integrity:** Validate all data types are correct. Name columns should be strings (str), continuous values should be floats (float64), and discrete values should be integers (int).
2. **Data Domain Validation:** Remove data is implausible such as negative diameter.
3. **Duplicate Data Removal:** Remove duplicate rows of data and observations of the same small-body. Multiple observations of the same small-body skew the sampling data in regression and create a data leakage problem with artificial accuracy in the case where an small-body observation ends up in the training and testing datasets.
4. **Target Variable Isolation (diameter):** All entries without a diameter have been removed. Diameter (Y) is the target variable and is required for the training model and to calculate the coefficient of determination ( $R^2_{test}$ ) during out-of-sample testing.
5. **Exclusion of High-Cost Features (albedo):** Albedo represents a Group 3 characteristic and was explicitly dropped from the dataset. Removing this feature ensures the model does not rely on cost-prohibited data and maintains operational cost-effective analysis. This also prevents data leakage between albedo and the target diameter.
6. **Feature Subset Selection:** The model requires the target diameter and only features that fall into Group 1 (e.g., *semi-major axis a*, *eccentricity e*, *inclination i*) and Group 2 (*absolute magnitude H*).
7. **Feature Completeness:** Entries with missing values in the selected feature subsets were removed. This ensures the regression algorithms receive a complete feature matrix and that any observed variance in  $R^2_{test}$  is attributable to specific feature subsets and not fluctuations in sample size or missing data.
8. **Target Variable Distribution Scaling:** Scale diameter (Y) using a base-10 logarithmic formula. Small-bodies size distribution is non-linear. Scaling the diameter (Y) ensures that the model weights small and large small-bodies the same.
9. **Outlier Detection:** Because outliers represent a real small-body and not an entry error or out-of-scope feature, outliers will be reviewed but may not be removed just on the nature of being an outlier.
10. **Feature Independence:** Remove any highly correlated features to reduce the risk of improper feature ranking between the correlated features.

# Analyze data

Data has been analyzed in the following ways:

1. **Individual Feature Boxplots:** Visualize distribution of individual features using boxplots.
2. **Target Normalization Check:** Visualize log transformation of target diameter (Y) using a histogram.
3. **Primary Signal Feature:** Visualize absolute magnitude (H) versus target diameter (Y) for correlation using a scatterplot.
4. **Orbital Characteristic Features:** Visualize features versus target diameter (Y), absolute magnitude (H), and other features for correlation and inclusion in model using 2D and 3D scatterplots with visual mapping.
5. **Excluded Features:** Determine features that do not require additional analysis.

# Report

## 1. Exploratory Data Analysis (EDA)

Preliminary analysis of the dataset shows that the key indicators of a small-body size (target diameter (Y)) are absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i). There is a strong non-linear relationship between diameter and H indicating that H is a strong predictor of diameter. While there were no linear correlations between individual a, e, i features and diameter or H, comparison of a, e, and i mapped to diameter and a, e, and i mapped to H provided similar results. This provides strong evidence that a, e, and i could meet the operational threshold to predict diameter without the need for H. The EDA alone is not enough to conclude if the null hypothesis or the alternative hypothesis are correct, but there is a pattern and relationship between a, e, i, and diameter.

## 2. Feature Determination

Analysis of the features indicates that a, e, and i are strong candidates for the regression model because they have a non-linear, structural relationship to the target. The feature q is reserved for future determination for inclusion or exclusion in the regression model since it is derived from a and e which could introduce multicollinearity or noise. Features om and w can be excluded. While om and w are not derived from any other features, the lack of correlation to the target may introduce noise to the regression model that could lead to overfitting.

## 3. Model Selection

An analysis of the dataset and feature determination support using a random forest regression model to test the hypothesis. Small-bodies form dense structural clustering in the feature space and a highly skewed target distribution. Small-bodies also can be observed at great distances which creates mathematical outliers. Physical characteristics and derived characteristics may have no correlation to the size of a small-body or introduce multicollinearity. A random forest regression model uses piecewise step-function approximations to handle extreme outliers and noise of derived or irrelevant features. This is better than linear regression or other regression models that try to fit data into a straight line or curve. This will result in stronger models to increase accuracy to predict small-body sizes and to test the hypothesis.

# Act

- <apply results, connect results with the chosen problem statement or business question>

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